

## Cognitive Companion — the ask: seed, not the whole envelope

A briefing document for the department chair. What we need from Anesthesiology, what we do not need, and who we need to be introduced to.

### The number that describes the work

An honest twelve-month build-and-test of Cognitive Companion — camera-free hardware, software engineering with the College of Engineering, a first simulation cycle, and a write-up that can travel — is about **\$155,000**.

That figure is the year-one **program envelope**. It is the size of the project. It is not the invoice we are handing the department.

### What we are actually requesting from the chair

Two things, in this order.

#### First: a substantial departmental seed of \$50,000.

We are not requesting \$155,000 from Anesthesiology. We are requesting enough money to stand the project up so the rest of the envelope can be raised from people whose job is to fund this kind of work.

The seed is a springboard. Without it, we cannot buy the kit, start engineering with Dr. Spínola's group, or run the first sim cycle, and every later conversation is still a slide. With it, the next ask is a working prototype plus process data plus an engineering partner who already said this is implementable.

#### Second: introductions.

We are asking the chair to help open doors we should not cold-call:

- 1 **The dean** — so this is visible as a School of Medicine educational and research bet, not a hobby in one unit.
- 2 **The School of Medicine research office or research group** that can provide more specific funds for early clinical-AI, simulation, or engineering-medicine work.
- 3 **Any other contact she trusts** — institute-level engineering-and-medicine leadership, simulation leadership, philanthropic internal funds, society-grant mentors.

The introductions are not a courtesy. For a \$50,000 seed to unlock a \$155,000 year, someone has to walk us to the next check.

### Why not ask the department for the full \$155,000

Anesthesiology should not be the entire bank for a build that is jointly clinical and software-engineering. The department should be the clinical home and the first believer. The College of Engineering is already in the conversation. The School of Medicine research apparatus is the right place for the balance. If we ask one chair to fund the whole envelope, we skip the people who can make this durable.

A seed also keeps the project honest. Fifty thousand dollars has to produce something visible in sim. That is a better discipline than a large internal earmark with no off-ramp.

## What \$50,000 of seed actually buys

Approximate, not a procurement list:

| Use                                                                                                                                        | Seed            |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Hardware kit (Halo and/or G2 in small quantity, dedicated iPad/iPhone, mounts, audio fallback)                                             | \$10,000        |
| Engineering start with College of Engineering (student or research-staff time, software-engineering standards, safety gate, protocol pack) | \$25,000        |
| First simulation cycle (scenario writing with faculty, session time, small learner incentives)                                             | \$10,000        |
| Contingency and compliance friction (IRB/QI determination, odds and ends)                                                                  | \$5,000         |
| <b>Departmental seed</b>                                                                                                                   | <b>\$50,000</b> |

Hardware is the small line. Engineering time is the real line. That is intentional.

## Where the rest of the \$155,000 should come from

Rough year-one envelope, so the chair can see we are not hiding the true cost:

| Use                                                                              | Year-one envelope |
|----------------------------------------------------------------------------------|-------------------|
| Software engineering and student/research staff (majority)                       | \$80,000          |
| Clinical and education effort (scenario ownership, session faculty, measurement) | \$25,000          |
| Hardware and dedicated devices                                                   | \$10,000          |
| Simulation operations beyond the first cycle                                     | \$12,000          |
| Compute, evidence licenses if any, basic infrastructure                          | \$8,000           |
| Dissemination (abstract, meeting, write-up)                                      | \$5,000           |
| Program contingency                                                              | \$15,000          |
| <b>Year-one program</b>                                                          | <b>\$155,000</b>  |

The seed covers hardware, the engineering start, and the first sim cycle. The remaining approximately \$105,000 is what we will take to the dean, to School of Medicine research, and to society mechanisms once the seed has produced an artifact.

We will not describe the NSF AIHealth-LL award as a piggy bank for this project. That award belongs to Dr. Spínola's living-lab education program with Dr. Falcão and Dr. Damevski. Its existence is evidence that VCU already funds AI-for-health engineering with clinicians. It is not a transfer.

## Industry money is last

We are evaluating the possibility of industry funds. We are treating that path as a **last resort**.

Industry money on a wearable or an AI cueing product creates a conflict of interest that is hard to wash out of simulation research, trainee education, and later clinical claims. A vendor who pays for the prototype will want the protocol layer, the story, or the access. We would rather arrive at any industry conversation with an internally seeded, camera-free, measured sim prototype than start there.

If the chair knows a clean philanthropic or foundation path, that is different from a device vendor. We will take that introduction. We will not lead with a company.

## What we need the chair to say yes to, in one pass

- 1 Anesthesiology will seed **\$50,000** toward Cognitive Companion.
- 2 The chair will introduce Navarrete to **the dean** and to **School of Medicine research leadership** (and anyone else she thinks is the right next dollar).
- 3 The department agrees the frame is **simulation and education first**, camera-free, no live-patient version 1, no industry-first funding.
- 4 The year-one success test is a working prototype plus process data plus a written springboard — not a press release and not an FDA claim.

## What we are not asking her to do

We are not asking her to certify clinical outcome benefit. We are not asking her to approve a wearable camera. We are not asking her to make this a live ICU device this year. We are not asking her to fund \$155,000 from the department. We are not asking her to broker a vendor deal.

## Closing

The clinical problem is real. The content evidence already exists. The packaging is what we have to build. The engineering partner on this campus has already reviewed the project and judged implementation to be substantially likely. We need a departmental seed so that judgment turns into software and sim data, and we need her network so Anesthesiology is the springboard rather than the entire bank.